
Problem 1. Conceptual Questions. Write an equation, an expression, or a short sentence to describe each of these concepts. [5 marks]

- (a) The line integral $\int_C \vec{F} \cdot d\vec{r}$ is **path-independent** in a domain D .

Solution.

□

- (b) The **exactness criteria** on $\vec{F}$ for the differential form $\vec{F} \cdot d\vec{r}$ to be an exact Pfaffian df for some scalar potential f .

Solution.

□

- (c) A vector field $\vec{F}$ is **incompressible**.

Solution.

□

- (d) A scalar field f is the **scalar potential** of a vector field $\vec{F}$.

Solution.

□

- (e) The **directional derivative** of a scalar field f in the direction of a unit vector $\hat{\mathbf{u}}$.

Solution.

□

Problem 2. Find $\vec{\nabla} f$ and its value at P for

$$f = \frac{x}{x^2 + y^2}, \quad P(1, 1).$$

[5 marks]

Solution.

□

Problem 3. Find the **divergence** of the following vector field $\vec{v}$ at the point $P(1, 0, -1)$:

$$\vec{v} = \frac{1}{(x^2 + y^2 + z^2)^{3/2}} (x\mathbf{i} + y\mathbf{j} + z\mathbf{k}).$$

[5 marks]

Solution.

□

Problem 4. Suppose $\vec{u} = u_1(x, y, z)\mathbf{i} + u_2(x, y, z)\mathbf{j} + u_3(x, y, z)\mathbf{k}$ and $\vec{v} = v_1(x, y, z)\mathbf{i} + v_2(x, y, z)\mathbf{j} + v_3(x, y, z)\mathbf{k}$. Prove the following formula:

$$\nabla \cdot (\vec{u} \times \vec{v}) = \vec{v} \cdot (\nabla \times \vec{u}) - \vec{u} \cdot (\nabla \times \vec{v}).$$

Hint: Use the triple scalar product formula:

$$\vec{a} \cdot (\vec{b} \times \vec{c}) = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}.$$

[5 marks]

Solution.

□

Problem 5. *Given that the differential form under the integral is exact (you don't need to show this), find the result of the following line-integral:*

$$\int_{(0,0,0)}^{(1,1,0)} e^{x^2+y^2+z^2} (x \, dx + y \, dy + z \, dz) .$$

[5 marks]

Solution.

□

Problem 6. *Calculate the area of the triangular region with vertices $(0, 0)$, $(0, 4)$, $(4, 4)$ using **only line integrals** along the boundary of the right-angled triangle.*

[5 marks]

Solution.

□